采样定理

$M(t) = \sum_{m=-\infty}^{\infty} \delta(t-mT)$

采样信号

$$X_a(j\Omega) = F[x_a(t)] = \int_{-\infty}^{\infty} x_a \cdot e^{-jt\Omega} dt$$

$$M(t) = \sum_{k=-\infty}^{\infty} A_k e^{jk\Omega_s t} \quad (\because M(t) \text{为周期函数})$$

then,
$$A_k = \frac{1}{T}\int_{-T/2}^{T/2} M(t) \cdot e^{-jk\Omega_s t} dt$$
$$= \frac{1}{T}\int_{-T/2}^{T/2} \sum_{m=-\infty}^{\infty} \delta(t-mT) e^{-jk\Omega_s t} dt$$
$$= \frac{1}{T}$$

$$\Rightarrow M(t) = \frac{1}{T}\sum_{k=-\infty}^{\infty} e^{jk\Omega_s t} \quad (\text{我们要} M(t) \text{的傅里叶级数})$$

$$M(j\Omega) = F[M(t)] = F\left[\frac{1}{T}\sum_{m=-\infty}^{\infty} e^{jk\Omega_s t}\right]$$
$$= \frac{1}{T}\sum_{k=-\infty}^{\infty} F(e^{jk\Omega_s t})$$
$$= \frac{1}{T}\sum_{k=-\infty}^{\infty} \left(2\pi\delta(\Omega - k\Omega_s)\right)$$
$$= \Omega_s \sum_{k=-\infty}^{\infty} \delta(\Omega - k\Omega_s)$$

采样函数的频域表示.

$$\hat{x}_a(t) = x_a(t)\cdot M(t), \quad \hat{X}_a(j\Omega) = FT[x_a(t)\cdot M(t)]$$

$$\Rightarrow \hat{X}_a(j\Omega) = \frac{1}{2\pi}[M(j\Omega) * X_a(j\Omega)]$$
$$= \frac{1}{2\pi}\cdot\Omega_s\cdot\int_{-\infty}^{\infty} X_a(j\theta)\cdot\sum_{k=-\infty}^{\infty}\delta(\Omega - k\Omega_s - \theta)d\theta$$
$$= \frac{1}{T}\cdot\sum_{k=-\infty}^{\infty}\int_{-\infty}^{\infty} X_a(j\theta)\cdot\delta(\Omega - k\Omega_s - \theta)d\theta$$
$$= \frac{1}{T}\cdot\sum_{k=-\infty}^{\infty} X_a(j\Omega - jk\Omega_s)$$